

### E.3 / PAPER 1 / QUESTION 28

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- 28.** A pure sample of radon-222 decays into polonium with a half-life of 4 days. A measurement is made of its radioactivity. The initial count rate is 420 counts per second. The measured background count is 20 counts per second.

What is the measured count rate after 8 days?

- A. 85
- B. 100
- C. 105
- D. 120